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2024_28_IV_CSD_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 8_Q1

Attempt : 2
Total Mark : 10
Marks Obtained : 7.5

Section 1 : Coding

1. Problem Statement

Write a program to validate the email address and display suitable exceptions if there is any mistake.

Create 3 custom exception classes as below

DotExceptionAtTheRateExceptionDomainException

A typical email address should have a " . " character, and a "@" character, and also the domain name should be valid. Valid domain names for practice be 'in', 'com', 'net', or 'biz'.

Display Invalid Dot usage, Invalid @ usage, or Invalid Domain message based on email id.

Get the email address from the user, validate the email by checking the above-mentioned criteria, and print the validity status of the input email address.

Input Format

The first line of input contains the email to be validated.

Output Format

The output prints a Valid email address or an Invalid email address along with the suitable exception

If email ends with . or contains not exactly one . after @, it throws:

DotException: Invalid Dot usage

Invalid email address

If @ appears not exactly once, it throws:

AtTheRateException: Invalid @ usage

Invalid email address

If the part after the last dot is not among accepted domains:

DomainException: Invalid Domain

Invalid email address

If all conditions satisfied then print:

Valid email address

Refer to the sample input and output for format specifications.

Sample Test Case

Input: sample@gmail.com
Output: Valid email address

Answer

```
// You are using Java
import java.util.Scanner;
class DotException extends Exception{
   DotException(String msg){
      super(msg);
   }

}
class AtTheRateException extends Exception{
   AtTheRateException(String msg){
      super(msg);
   }
}
class DomainException extends Exception{
   DomainException(String msg){
      super(msg);
   }
}
class demo{
   static void chkEmail(String email) throws AtTheRateException, DotException ,
DomainException{
      int a = email.replace("@","").length();

      if(email.length()-a!=1){
         throw new AtTheRateException("Invalid @ usage");
      }
      if(email.endsWith(".")){
         throw new DotException("DotException");
      }
      int indAt = email.indexOf('@');
      String aftrAt = email.substring(indAt+1);
      int b = aftrAt.length()-aftrAt.replace(".","").length();
```

```java
        if(b!=1){
            throw new DotException("DotException");
        }
        int indDot = email.indexOf(".");
        String domain = email.substring(indDot+1);
        if(domain.equals("in")|| domain.equals("com")|| domain.equals("net")||
domain.equals("biz")){
            System.out.println("Valid email address");
        }else{
            throw new DomainException("DomainException");
        }

    }
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        try{
            String email = sc.nextLine();
            chkEmail(email);
            //System.out.println("Valid email address");
        }
        catch(DotException e){
            System.out.println(e.getMessage() +": Invalid Dot usage");
            System.out.println("Invalid email address");
        }
        catch(AtTheRateException e){
            System.out.println(e.getMessage()+": Invalid @ usage");
            System.out.println("Invalid email address");
        }
        catch(DomainException e){
            System.out.println(e.getMessage()+": Invalid Domain");
            System.out.println("Invalid email address");
        }


    }
}
```

Status : Partially correct ***Marks : 7.5/10***